Calculus II

MTH 231 | Section C | Fall 2012

Dr. James Ashe

Johnson C. Smith

Calc II: 4.8 Antiderivaties

Review

Using limits to avoid division by zero

A ball rolls down a ramp so that its height h from the top after t seconds is t^2. What is the instantaneous rate of change of the h with respect to t after 3 seconds?

- i.e., what is its velocity at $t = 3$.

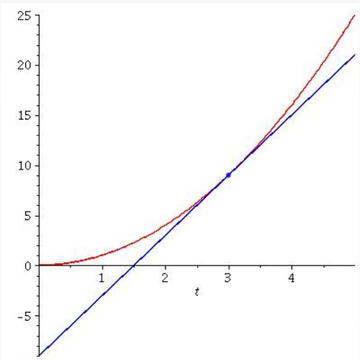
$$\frac{\triangle h}{\triangle t} = \frac{(4)^2 - 3^2}{4-3} = 7 \text{ ft/sec}$$

$$\frac{\triangle h}{\triangle t} = \frac{(3.25) - 3^2}{3.25-3} = 6.25 \text{ ft/sec}$$

$$\vdots$$

$$\lim_{t\to 3}\frac{\triangle h}{\triangle t} = \lim_{t\to 3}\frac{t^2-3^2}{t-3} = \lim_{t\to 3}\frac{(t-3)(t+3)}{(t-3)}$$

$$\lim_{t\to 3}\frac{\triangle h}{\triangle t} = \lim_{t\to 3} t+3 = 6$$

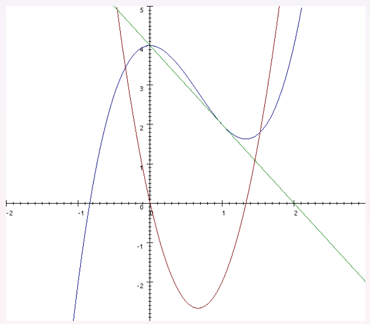


Notation: The derivative of a function, $f(x)$

- The derivative of f with respect to x written: $f'(x) = \frac{d}{dx}f(x) = \frac{df(x)}{dx} = \frac{df}{dx}$.
- The derivative of f with respect to x *at a value "a"*: $f'(a) = \frac{d}{dx}f(a)$.
  - This is the "slope" of f at a.

Example

- $f(x) = 2x^3 - 4x^2 + 4 \Rightarrow$ $f'(x) = 6x^2 - 8x$
- For a =1, $f'(1) = 6(1)^2 - 8(1) = -2$



Questions

1) Suppose you are a driving at a constant rate of 50 $^{mi}/_{hr}$. How far do you travel in the first 3 hours?

   $50 \times 3 = 150$

2) Suppose a ball rolls down a ramp at velocity of $2t$ $^{ft}/_{sec}$ where t is the number of seconds after it started to roll. How far does the ball travel during the first 3 seconds?

   Here the velocity varies according to time. Clearly, the derivative is related to this problem, but instantaneous velocity occurs instantly. This problem will again require limits, but instead of tangent lines we will find areas.

Antiderivatives

Definition

A function F is an ***antiderivative*** of f on an interval I if and only if $F'(x) = f(x)$ for all $x \in I$.
Find all the antiderivatives of the following functions

ex 10) $g(x) = 11x^{10} \Rightarrow G(x) = x^{11} + C$ where C is some constant number.

Note that: $\frac{d}{dx}(x^{11} + 2) = \frac{d}{dx}(x^{11} + \pi) = \frac{d}{dx}(x^{11} + \frac{1}{2}) \cdots = 10x^{11}$.
So $x^{11} + C$ represents the *family* of antiderivatives of f.

Notation

$\int f(x)dx$, the *indefinite integral*, represents the family of antiderivatives of $f(x)$.

If $F(x) = \frac{x^{p+1}}{p+1} + C$ then $F'(x) = (p+1)\frac{1}{p+1}x^{p+1-1} = x^p$.
So then $F'(x) = x^p = f(x)$.

Theorems:

Power rule: $\int x^p dx = \frac{x^{P+1}}{p+1} + C$ where $p \neq -1$ is a real number.
Sum rule: $\int (f(x) + g(x))dx = \int f(x)dx + \int g(x)dx$
Constant rule: $\int cf(x)dx = c\int f(x)dx$

ex) $\int (-6\frac{1}{z^7} + 2)dz = -6\int z^{-7}dz + \int 2dz$
$= z^{-6} + 2z + C$

Recall the chain rule, $(f \circ g)'(x) = f'(g(x))g'(x)$.

- So then $\frac{d}{dx}(\sin ax) = a \cdot \cos(ax)$
- Similarly we can find the indefinite integrals of trigonometric functions; see Table 4.5 pg. 243.

Introduction to Differential Equations

A ball rolls down a ramp so that its velocity is $v(t) = 2t$ feet per second. The distance of the ball from the top of the ramp is 3 feet after 1 second. Find the function giving the distance of the ball from the top of the ramp at t seconds.

$\int 2t dt = t^2 + C$; an infinite class of solutions.

$h(t) = t^2 + C \Rightarrow h(1) = 1 + C = 3 \Rightarrow C = 2$

So $h(t) = t^2 + 2$.

HW: 15, 19, 24, 30, 38, 58

Next time, 5.1: Approximating areas under curves